

Whole-Brain Models of Minimal Phenomenal Experience: Approaching Criticality through Jhāna Meditation

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Abstract

Minimal phenomenal experiences (MPE) provide a distinctive entry point for investigating consciousness and understanding how certain meditative practices may represent and contribute to human flourishing. Adopting a neurophenomenological stance, we combine first-person phenomenology with formal whole-brain modeling to ask what mechanisms underlie MPE states and the progression towards it, as reached through jhāna meditation, a type of advanced concentrative absorption meditation (ACAM-J). We used a dataset of advanced practitioners performing the eight ACAM-J states scanned in high-field functional magnetic resonance imaging. For each state, we build a mechanistic whole-brain model that reproduces brain activity by modeling the dynamical regimes of different resting-state networks. We found that the later ACAM-J states, taken here as candidates for MPE, show a shift of functional network dynamics toward near-bifurcation working points. The default mode network exhibits the largest shift, from a distant noise-driven regime at a control condition to near-critical dynamics. We also observed that the pathway to MPE is non-linear, with prominent reconfigurations at key meditative milestones. Our results suggest that MPE corresponds to a globally susceptible state where near-critical dynamics dominate. We interpret this critical regime as a form of “openness”, where the usual hierarchical and heterogeneous patterns of brain activity give way to generalized flexibility. In this light, MPE provides both a minimal model of consciousness and a window into how advanced meditation can profoundly reshape brain dynamics.

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Introduction

Minimal phenomenal experience and contemplative science

A central challenge in the science of consciousness is understanding the minimal conditions for conscious experience. This question has recently received scientific attention under the framework of minimal phenomenal experience (MPE) (Metzinger 2020), which aims to identify the core causal factors for consciousness to occur. A central case for MPE is the phenomenon of *pure awareness*, with a recent compilation of over 500 phenomenological reports on the topic (Metzinger 2024). Despite the heterogeneity across traditions and practitioners, a core recurring phenomenology emerges from these reports: a state of presence with minimal phenomenal content, no egoic identification, and a distinctive quality of luminosity or “knowing”. These experiences are scientifically significant because they suggest that features often assumed to be necessary for consciousness, such as a self-model or temporal representation, may not be required.

Contemplative science provides a unique opportunity to investigate MPEs. Indeed, the neurophenomenological approach introduced by Varela emphasizes the need to combine first- and third-person perspectives for a scientific understanding of consciousness (Varela 1996). Within this field, a third wave of meditation research has focused on advanced practices that unfold with time and mastery (Sacchet et al. 2024). Beyond scientific exploration, MPEs also have broad implications for human flourishing, as many contemplative traditions regard them as pathways to liberation and release from suffering (Sparby et al. 2025).

The jhānas as a paradigm of minimal experience

As a potential study topic in this context, we turn to *jhāna* practice, a type of advanced absorption meditation. The jhānas are a series of eight sequential meditative absorption states recognized in most Buddhist traditions, although their descriptions vary across lineages. Contemporary practice manuals provide modern accounts of these states (Brahm et al. 2006; Brasington 2015; Burbea 2014; Ingram 2018), and a more systematic classification of the jhānas as a type of advanced concentrative absorption meditation (ACAM-J) was recently proposed (Sparby et al. 2024). These states have relatively well-defined phenomenological qualities, and a sharp distinction is usually drawn between the *form* (the first to fourth) and the *formless* (the fifth to eighth) ACAM-J. The main difference between these two categories is that the perception of physical form and sensory objects generally vanishes or becomes attenuated in the formless ACAM-J (Sparby et al. 2024).

The formless ACAM-J offer a particularly suitable case for studying minimal phenomenal experience, as they appear to involve a gradual deepening toward minimal content. The fifth jhāna is typically described as an immersion in boundless space, experienced as infinite, empty, or immeasurable, and accompanied by profound equanimity (Brahm et al. 2006; Brasington 2015). In the sixth jhāna, the focus shifts to boundless consciousness, sometimes described as awareness aware of itself, vast and limitless (Brasington 2015; Ingram 2018). The seventh jhāna is characterized by nothingness or voidness, often reported as an experiential absence that can

evoke both peacefulness and, at times, subtle fear (Brasington 2015; Ingram 2018). Finally, the eighth jhāna, described paradoxically as neither perception nor non-perception, involves such a radical attenuation of experience that only the faintest trace of perception remains, making it difficult to describe in consistent terms (Brasington 2015; Ingram 2018). This progression may, in some cases, culminate in extended *cessations*, in which consciousness is said to “turn off” (Burbea 2014; Laukkonen et al. 2023; Yang et al. 2025).

This phenomenological progression shows that, across these stages, sensory impressions fade, subject–object structure weakens, and experiential content becomes increasingly attenuated. Interestingly, these phenomenological features closely converge with accounts of *pure awareness* collected in Metzinger’s recent compilation (Metzinger 2024). In this study, we therefore use the formless ACAM-J as a paradigmatic example of minimal phenomenal experience. Furthermore, because the progression through these states is relatively structured, it provides a unique opportunity to study the mechanisms underlying the gradual deepening toward MPE.

Beyond their scientific relevance, pure awareness and related phenomena may also be approached as transformative experiences. Within the Buddhist tradition, the role of the jhānas has been interpreted in different ways: some regard them primarily as supports for the cultivation of insight, others as sufficient on their own for approaching liberation, and still others describe a close interconnection between absorption and insight at each stage (Anālayo 2016; Burbea 2014). As an example, the *Anupada Sutta* in the Pali canon, one of the earliest records of Buddhist teachings, illustrates the integration of absorption and insight:

Furthermore, with the complete transcending of the dimension of nothingness, Sariputta entered and remained in the dimension of neither perception nor non-perception. He emerged mindfully from that attainment. On emerging mindfully from that attainment, he regarded the past qualities that had ceased and changed: ‘So this is how these qualities, not having been, come into play. Having been, they vanish.’ He remained unattracted and unrepelled with regard to those qualities, independent, detached, released, dissociated, with an awareness rid of barriers. He discerned that ‘There is a further escape,’ and pursuing it there really *was* for him. (MN 111) (Bhikkhu 2007)

Only a few empirical studies have investigated ACAM-J in laboratory settings. A recent single-subject 7T fMRI case study found state-specific whole-brain regional homogeneity patterns that tracked the ACAM-J sequence, with anterior–posterior reorganization and correlations with first-person ratings (Yang et al. 2024), effects that proved robust across repeated fMRI runs in the same participant (Ganesan et al. 2024). The same dataset has been analyzed using functional gradients and geometric eigenmodes, showing a flattened principal gradient and increased differentiation of sensory and attention regions (Demir et al. 2025). This change has been alongside increased global power in ACAM-J and an evolution in mid-spectrum modes across ACAM-J states that aligns with reported phenomenology (Potash et al. 2025).

The neurophenomenological approach started by Varela involves not only integrating first- and third-person data, but also developing formal models that can bridge those two (Lutz et al. 2025; Thompson et al. 2005). In recent years, several studies have applied computational accounts to

link formal theories with phenomenology and neurobiological data in the context of meditation, most often using active inference frameworks (Laukkonen et al. 2021; Sandved-Smith et al. 2021; Tal et al. 2025). A few recent papers have focused specifically on jhāna meditation, converging on the idea that these practices involve a reduction of hierarchical modeling (Lopez-Sola et al. 2025; Mago et al. 2024; Prest et al. 2024). In these accounts, the progression through the jhānas is described in terms of a phenomenology of “letting go,” modeled as a progressive reduction of prior precision or as the dissolution of attractor states along the modeling hierarchy. These models have also been linked to empirical predictions, such as increased entropy and reduced alpha power in deeper jhāna states, reflecting loosened top-down constraints.

Whole-brain modeling as a bridge

Although such recent methodological accounts have provided useful frameworks, they remain far from the underlying mechanisms through which the brain reorganizes. It is difficult to interpret the growing body of fMRI and EEG findings on deep meditative absorption states without models that reproduce the underlying brain mechanisms. Whole-brain models may offer such a bridge. These models simulate large-scale brain activity by combining mathematical descriptions of local dynamics, (from abstract mathematical models reflecting different activity regimes to neural mass or mean-field models describing the underlying neurophysiology), with empirical anatomical constraints, typically connectivity patterns derived from diffusion MRI and tractography (Deco et al. 2013). By adjusting local or global parameters, whole-brain models can reproduce a wide range of empirical phenomena, including resting-state functional connectivity, metastability, and signal complexity (Breakspear 2017; Deco et al. 2017).

Whole-brain models have already been applied to a variety of states of consciousness, including sleep, anesthesia, and disorders of consciousness, providing a mechanistic account for the large-scale reorganization of brain dynamics observed in these conditions (Deco et al. 2018a; López-González et al. 2021; Luppi et al. 2022; Mindlin et al. 2024; Sanz Perl et al. 2021). They have also been used to study pharmacologically-induced states such as those produced by serotonergic psychedelics, where whole-brain models help capture the characteristic increase in signal entropy and flattening of functional hierarchy (Deco et al. 2018b; Gendra et al. 2024; Herzog et al. 2023; Vohryzek et al. 2024b). These effects are consistent with the REBUS model, which interprets them as a relaxation of high-level priors and a reduction of top-down constraints, and with the entropic brain hypothesis, which links such changes to greater signal complexity and entropy (Carhart-Harris 2018; Carhart-Harris et al. 2019; Ruffini et al. 2023). Such studies are especially relevant for the current study, since, as mentioned above, deep absorption practices like ACAM-J have also been described as a progressive loosening of hierarchical predictions accompanied by heightened entropy.

Related to such frameworks is the notion of *criticality*. Empirical and modeling studies suggest that the brain operates near a critical point, allowing efficient computation, flexible adaptation, and memory preservation (Chialvo 2010; Cocchi et al. 2017). At this regime, small perturbations can propagate across the system, amplifying their effects and increasing complexity, while still maintaining enough stability to support a coherent function. Whole-brain models have repeatedly shown that the best fits to empirical data are obtained when model parameters lie

close to such critical values, striking a balance between order and instability that supports both consistency and adaptability (Deco et al. 2012; Vohryzek et al. 2022). Computational modeling thus provides a powerful way to study criticality, which is a key framework for understanding how different conscious states, including deep meditative absorption, reorganize large-scale brain dynamics.

Only two studies have so far applied large-scale models to meditative states (Dagnino et al. 2024; De Filippi et al. 2022). Both examined vipassanā *anapanasati* practice, which combines concentration and insight but lacks the absorption qualities that make the deeper ACAM-J states strong candidates for minimal phenomenal experience. To our knowledge, the present study is the first to investigate advanced meditative absorption states with whole-brain models, offering a unique opportunity to better understand these states and the mechanisms through which such minimal experiences emerge.

Our study aims to investigate the neural mechanisms underlying minimal phenomenal experiences by providing a mechanistic account of the alterations observed in fMRI data during advanced concentrative absorption meditation states. Specifically, we explore what causal mechanisms give rise to such experiences and how they unfold in the progression towards them. This was done by fitting whole-brain models to fMRI data from 14 subjects across eight ACAM-J states and a control condition, with the models parametrized by seven resting-state networks (RSNs) to assess their mechanistic involvement. In practical terms, this modeling framework allowed us to address our two central questions by testing (i) how the causal involvement of different RSNs can account for brain activity in the deepest ACAM-J states, particularly the formless ACAM-J and J7–J8, and (ii) how changes in RSN involvement may mechanistically explain the progression through the ACAM-J states towards MPE.

Methods

Participants

The present dataset included 17 advanced ACAM-J practitioners, extending an earlier case study that had examined a single adept meditator (Yang et al. 2024). We removed three participants from the analysis as two participants did not have fMRI recording from either J7 and/or J8, and one participant had only seven time points in J8, making the estimation of functional connectivity challenging. All participants provided informed consent, and ethical approval was obtained from the relevant institutional review board. Similar to the original study, participants were screened for neuropsychiatric and cognitive health to ensure suitability for high-field fMRI research.

Meditation Protocol

Each participant performed a standardized sequence of ACAM-J inside the MRI scanner. The sequence involved entering access concentration (Brasington 2015), progressing systematically through the first to eighth ACAM-J (J1–J8), and concluding with a post-jhānic “afterglow” state. As in the case study (Yang et al. 2024), transitions into each ACAM-J state were

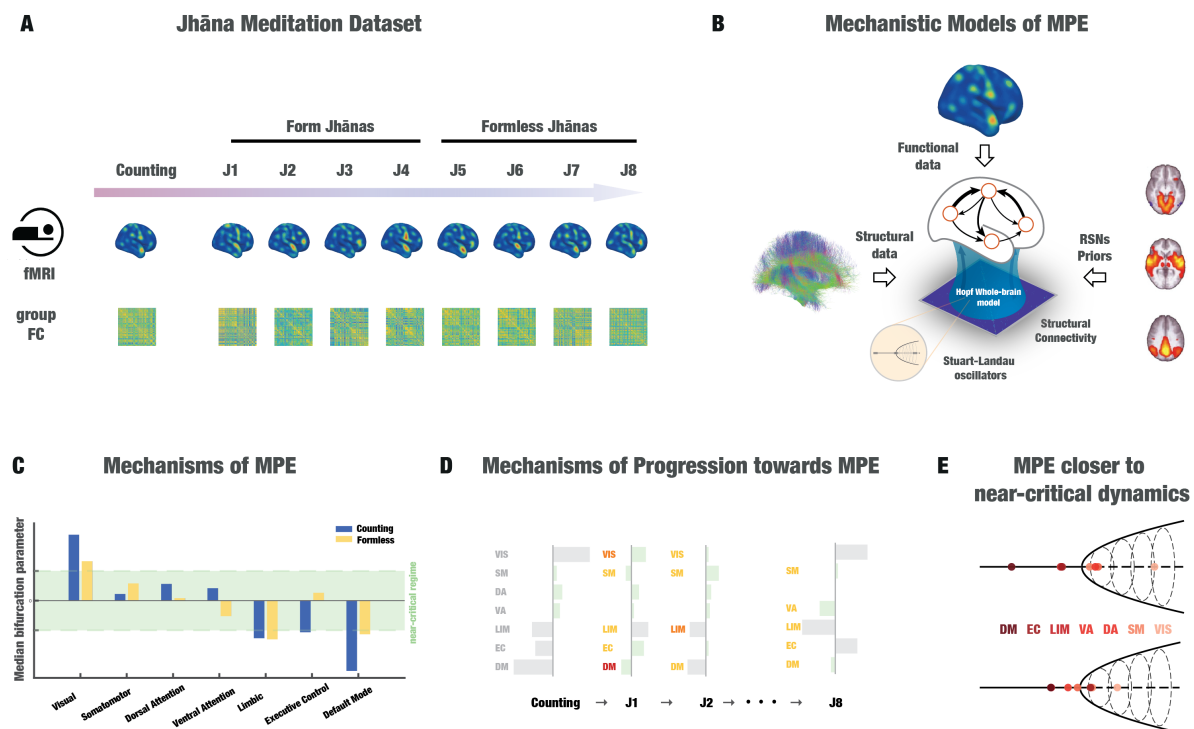


Figure 1: Investigation of mechanisms driving MPE in ACAM-J. **A:** Dataset description: we used fMRI data of 14 subjects undergoing ACAM-J. We computed a group functional connectivity (FC) to fit whole-brain models for each of the 8 ACAM-J states and the control counting condition. **B:** We created whole-brain models to fit the group FC. Whole-brain models, based on coupled local dynamics reflecting Stuart-Landau oscillators and constrained by a normative structural connectome, were fitted to the group FC of each condition. We used the resting-state networks defined in Yeo et al. (2011) as spatial priors for a genetic algorithm that fitted the bifurcation parameter of each local oscillator. **C:** We found that for most RSNs the bifurcation parameter converged towards near-bifurcation dynamics in MPE (here represented by the formless ACAM-J). **D:** We analyzed the evolution of dynamical regimes across ACAM-J states, finding stronger changes in the transition to J1 and the passage from the form to the formless ACAM-J, and a smoother progression throughout the formless ACAM-J. **E:** We discuss the hypothesis that MPE is characterized by a collapse of the brain’s dynamical regimes towards near-critical dynamics.

indicated using button presses.

Control Conditions

To provide non-meditative comparison conditions, participants completed two tasks designed to prevent inadvertent entry into meditative absorption: *Memory recall* (mentally recounting personal events from the previous two weeks) and *Mental counting* (silently counting backward in decrements of five from a large starting number, e.g., 10,000). Each control task lasted approximately 8 minutes. Runs were scheduled both before and during meditation sessions, as in the original design (Yang et al. 2024). For our analysis, we focused on the counting condition as a control for meditative practitioners.

Neuroimaging Acquisition and Preprocessing

MRI data acquisition parameters followed the case-study protocol (Yang et al. 2024): high-resolution 7T fMRI (SIEMENS MAGNETOM Terra, 32-channel head coil) with whole-brain coverage, along with structural T1-weighted scans. Physiological signals (heart rate, respiration) were recorded concurrently to allow correction for physiological noise.

Data preprocessing steps replicated those from the single-case design, including de-spiking and physiological noise regression (RETROICOR), slice-timing and distortion correction, motion correction and scrubbing, normalization to standard space, band-pass filtering (0.01–0.1 Hz), spatial smoothing (2mm FWHM), and z-scoring of time series. Each run was segmented into ACAM-J states (J1–J8) or control-task periods, yielding condition-specific time series for each subject.

Structural Connectivity

The $N \times N$ matrices of structural connectivity (C) used for the network model were derived from a probabilistic tractography-based normative connectome provided as part of the leadDBS toolbox (Horn et al. 2015). This normative connectome was generated from diffusion-weighted and T2-weighted Magnetic Resonance Imaging (MRI) from 32 healthy participants (mean age 31.5 years, 14 females) from the Human Connectome Project (HCP). The diffusion-weighted MRI data were recorded for 89 minutes on a specially designed MRI scanner with more powerful gradients than conventional scanners. The dataset and acquisition protocol details are available in the Image & Data Archive under the HCP project (<https://ida.loni.usc.edu/>). DSI Studio (Yeh 2025) was used to implement a generalized q-sampling imaging algorithm on the diffusion data. A white-matter mask, derived from the segmentation of the T2-weighted anatomical images, was used to co-register the images to the b0 image of the diffusion data using the SPM12 toolbox (Ashburner et al. 2014). Within the white-matter mask, 200,000 most probable fibres were sampled for each participant. The fibres were then transformed to the standard Montreal Neurological Institute (MNI) space using a nonlinear deformation field derived from the T2-weighted images via a diffeomorphic registration algorithm. The individual tractograms were aggregated into a joint dataset in MNI standard space, resulting in a normative tractogram representative of a healthy young adult population and made available in the leadDBS toolbox. The $N \times N$ matrices were computed from the normative tractogram using the Schaefer-200 parcellation scheme (Schaefer et al. 2018) with $N = 200$ cortical parcels. Structural connectivity values $C(n, p)$ were calculated as the number of tracts between each pair of brain areas n and p .

Whole-Brain Model

We implemented a Hopf bifurcation model to describe local neural mass dynamics. Each brain region j was modeled as:

$$\frac{dz_j}{dt} = (a_j + i\omega_j)z_j - |z_j|^2 z_j + G \sum_k C_{jk}(z_k - z_j) + \beta \eta_j(t)$$

where $z_j = x_j + iy_j$ represents regional activity, a_j is the bifurcation parameter, ω_j the intrinsic frequency (estimated from fMRI peak frequency), C_{jk} the structural connectivity weight, G a global coupling parameter, and $\beta\eta_j$ Gaussian noise. The bifurcation parameter represents different dynamic regimes of the Hopf oscillator — $a_j < 0$: stable fixed-point (noise-driven), $a_j > 0$: self-sustained oscillations, $a_j \approx 0$: complex, metastable dynamics.

Optimization

In order to mechanistically test the influence of different resting-state networks in ACAM-J and subsequently MPE, regional bifurcation parameters were grouped using spatial priors to reflect seven canonical resting-state networks (Yeo et al. 2011)—Visual (VIS), Somatomotor (SM), Dorsal Attention (DA), Ventral Attention (VA), Limbic (LIM), Executive Control (EC), and Default Mode (DM).

First, we fixed the global coupling to $G = 0.5$, as optimized in previous literature (Ipiña et al. 2020). Spatial heterogeneity in local dynamics was introduced by grouping brain regions according to anatomically motivated priors (in this case, the mapping of the seven resting-state networks into the Schaefer-200 parcellation), with each group contributing an independent coefficient to the bifurcation parameter of its nodes. These coefficients were estimated using a genetic algorithm that iteratively improved candidate solutions based on their fit to the empirical data. The empirical observables were derived from averaging participants per condition (across fourteen subjects), producing group-level FC matrices (calculated in terms of pairwise Pearson correlation) for each ACAM-J. Fitness was quantified as one minus the structural similarity index (SSIM) between empirical and simulated connectivity matrices.

Each optimization run began with a population of 10 individuals with random bifurcation parameter values close to zero (centered around $a = -0.01$) and progressed through cycles of elite retention, crossover, and mutation until convergence criteria were met (maximum 20 generations or stabilization of the best solution). In each generation, 20% of individuals were preserved through elite selection, 60% were produced by crossover of parent solutions, and 20% by mutation. Forty independent runs were performed for each ACAM-J state, and the resulting distributions of parameters were used to assess stability.

We also ran a homogeneous optimization, where we optimized a global bifurcation parameter. Unlike the heterogeneous case, where we parametrized the bifurcation parameter of each RSN, here we optimized only one bifurcation parameter.

Statistical Analysis

To contrast model outputs across conditions, we generated forty independent realizations and derived the corresponding distributions of values. Differences between conditions were quantified using Cohen’s d , which captures the magnitude of effect size. As in previous work (Perl et al. 2023), we do not report p-values or other significance metrics dependent on sample size, since the apparent statistical power could be inflated arbitrarily by increasing the number of model realizations.

Results

To investigate the mechanisms underlying the progression into MPE, we first analyzed fMRI BOLD data from 14 advanced practitioners scanned during the counting control condition and across the eight successive ACAM-J states (Figure 1A). For each condition, we computed the group-level functional connectivity and then fitted a whole-brain model of coupled Stuart–Landau oscillators (Hopf model), constrained by normative structural connectivity and informed by priors from the seven canonical resting-state networks (Figure 1B). Using a genetic algorithm, we optimized the bifurcation parameters of the model to reproduce the empirical functional connectivity patterns for each state, yielding network-level dynamical regimes across counting and ACAM-J states. This parametrization allowed us to characterize the dynamical regime of each resting-state network in relation to the bifurcation point (Figure 1C). By comparing these network-level bifurcation parameters across the ordered progression of ACAM-J states, we identified characteristic trajectories of RSN dynamical shifts (Figure 1D). Finally, we discuss the implications of these converging network dynamics for understanding how the brain approaches criticality as practitioners progress toward MPE (Figure 1E).

Changes in dynamical regimes in MPE

To characterize the dynamical regimes of each resting-state network across conditions, we examined the bifurcation parameters estimated through genetic algorithm optimization of the whole-brain models. By comparing the control (*counting*) condition with the formless ACAM-J (J5–J8) as states representative of MPE, we identified systematic changes in how RSNs approach or move across the Hopf bifurcation, revealing large-scale reconfigurations in the dynamical regimes different networks.

In the counting condition, RSNs showed clear differences in dynamical regimes and in the bifurcation parameter distance from the transition point (Figure 2). Several networks, including the Visual, Limbic, Executive Control, and Default Mode, were relatively far from the bifurcation, situated deep in the oscillatory (VIS) and noise-driven (LIM, EC, DM) regimes. In contrast, the Sensorimotor, Dorsal Attention, and Salience networks started much closer to the bifurcation, with dynamics already poised near the transition between oscillatory and noise-driven activity.

When comparing the formless ACAM-J to the control (counting) condition, many RSNs exhibited important changes, and these changes were generally in the direction of approaching the bifurcation (Figure 2B). The Visual network and the Default Mode network shifted closer to the bifurcation point with moderate effect sizes ($d > 0.5$). The Executive Control network and the Dorsal Attention network also moved towards the transition point, but with weak effect sizes ($d = -0.40$ and $d = 0.26$, respectively). The Ventral Attention network shifted from oscillatory toward noise-driven dynamics with a moderate effect size ($d = 0.33$), but remained close to the bifurcation across both conditions. The Somatomotor and Limbic networks exhibited negligible effect sizes ($d = -0.06$ and $d = 0.10$, respectively). Importantly, this trend was replicated when using the last formless ACAM-J (J7 and J8) as candidate MPE states, confirming the robustness of these findings (see Supplementary Figure S1). In summary, the shift toward

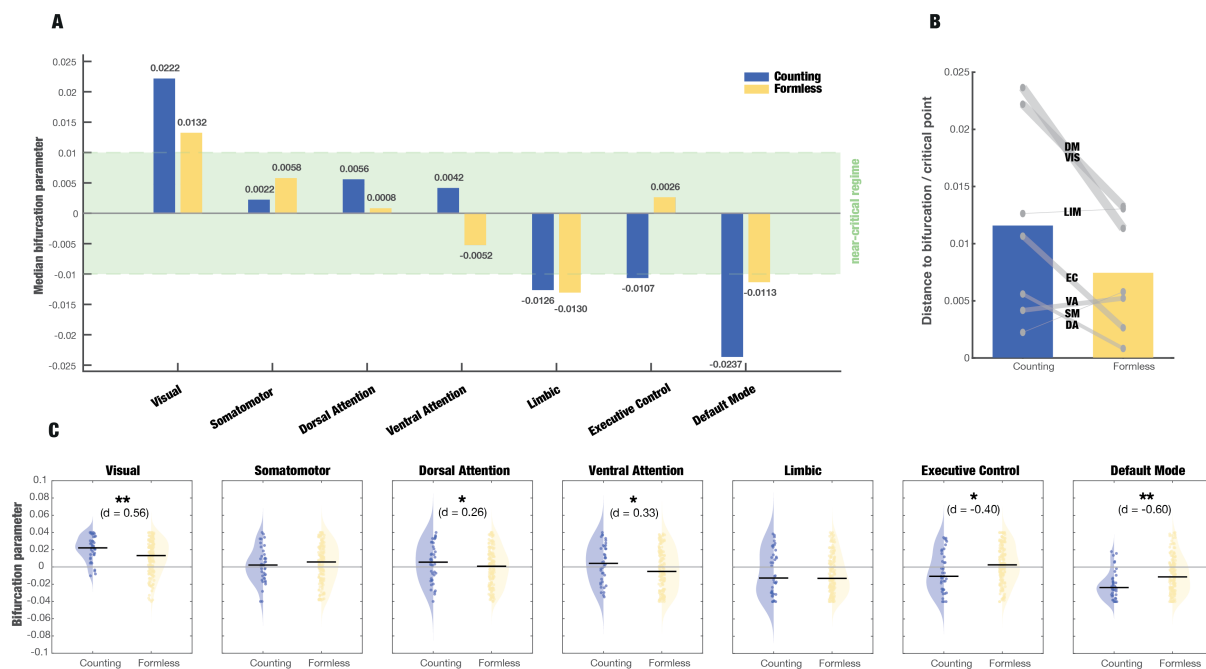


Figure 2: MPE shifts network dynamical regimes towards near-bifurcation dynamics. Bifurcation parameter values per RSN in the counting condition and the formless ACAM-J. **A:** Median bifurcation parameter values per RSN in the counting and formless conditions. The green shaded area marks the near-critical region ($|a| < 0.01$). **B:** Median distance of RSNs to the bifurcation point ($a = 0$). Dots indicate median distances per RSN, and line width represents the effect size according to Cohen's d . **C:** Distribution of bifurcation parameters per RSN for forty runs of the genetic algorithm in each condition. Solid lines indicate median values, * and ** indicate weak ($0.2 < |d| < 0.5$) and moderate ($0.5 < |d| < 0.8$) effect sizes according to Cohen's d .

the bifurcation was evident across several networks, but its magnitude and direction varied, reflecting heterogeneous trajectories.

In the control condition, the Sensorimotor network was the closest to the bifurcation, with dynamics already tuned near the bifurcation (median $a = 0.0022$). In the formless ACAM-J, the SM remained near the bifurcation, but the Dorsal Attention network became the closest to the transition point (median $a = 0.0008$, see Figure 2A). Among all RSNs, the Default Mode network underwent the strongest shift between conditions. In the control condition, the DM was relatively far from the bifurcation with a negative value of the bifurcation parameter, indicating that its dynamics were deep in the noise-driven regime (median $a = -0.024$). However, in the formless ACAM-J the DM was markedly closer to the bifurcation, showing the largest effect size across all networks ($d = -0.60$). We will comment on a potential interpretation of such shifts in the discussion.

The progression towards MPE

To investigate the brain mechanisms involved in the progression towards the MPE state, we examined the progression of RSNs parameters across ACAM-J states step by step (Figure 3). The results reveal that the approximation towards near-bifurcation dynamics is not linear but unfolds through a sequence of intermediate reconfigurations.

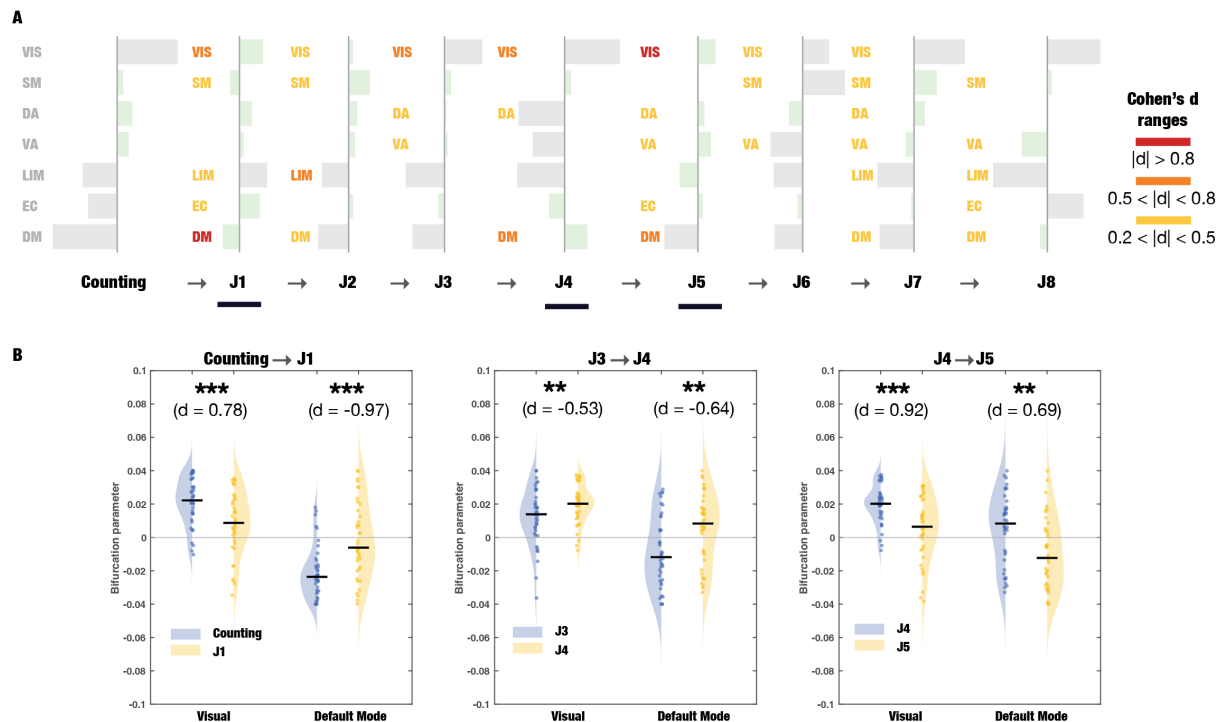


Figure 3: Mechanistic progression towards MPE. A: Progression of the bifurcation parameter in each RSN across the different ACAM-J states. The trajectory toward MPE is marked by key transition points (solid black line) in the transitions to J1, J4, and J5, after which the changes in dynamics are smoother. Bars indicate median values, green bars indicate proximity to bifurcation point ($|a| < 0.01$). Colors indicate the Cohen's d effect size in the change from the previous condition. **B:** Distribution of bifurcation parameters for the Visual and Default Mode networks in the three key transitions. Solid lines indicate median values, ** and *** indicate moderate ($0.5 < |d| < 0.8$) and strong ($|d| > 0.8$) effect sizes according to Cohen's d.

When focusing on moderate-to-strong effects ($|d| > 0.5$), the Default Mode and Visual networks emerged as the most mechanistically involved (Figure 3A). Interestingly, those were the networks furthest from the bifurcation point in the control condition. Three transition points stood out as particularly important. First, the comparison between the control condition and J1 already produced strong reconfigurations: the VIS (approaching from the oscillatory side) and the DM (from the noise-driven side) both moved markedly closer to the bifurcation with moderate and strong effect sizes, respectively. Second, in the transition to J4 (the peak of the form ACAM-J), the VIS and DM again showed pronounced shifts. In this case, the DM moved closer to the bifurcation, crossing from a noise-driven to an oscillatory regime ($|d| > 0.5$), while the VIS moved further from it within the oscillatory regime ($|d| > 0.5$). Third, in the transition from J4 to J5 (the passage from form to formless ACAM-J), large changes were again observed in both DM and VIS, marking another key reconfiguration. Here, the VIS approached the bifurcation ($|d| > 0.8$), whereas the DM did not and instead reverted back toward the noise-driven regime ($|d| > 0.5$).

The progression through the formless ACAM-J (J5–J8) was characterized by smaller effect sizes. The general trend was one of broad convergence toward the bifurcation, but this was not uniform. Both DM and SM networks continued to make modest but consistent shifts toward

the transition point ($|d| > 0.2$ in most cases). EC, DAN, and SAL networks remained very close to the bifurcation across the formless stages, with only the EC network diverging away at J8. In contrast, the LIM and VIS networks progressively moved away from the bifurcation, with the LIM network shifting deeper into the noise-driven regime and the VIS network drifting further into the oscillatory regime. By J8, the LIM network was the furthest from the bifurcation, followed by the VIS network (although it remained closer than it had been at the control condition, as shown by Figure 2).

Together, these results indicate a trajectory toward MPE marked by a few key transition points, i.e., early engagement at J1, a major reconfiguration at the last form ACAM-J (J4), and another at the passage into the formless ACAM-J (J5), after which the system stabilizes into a broadly, but heterogeneously, near-bifurcation regime. We confirmed this pattern by comparing the bifurcation parameters obtained from whole-brain model fittings of the average FC for the form and formless ACAM-J (see Supplementary Figure S2). As above, this analysis revealed a major transition toward near-bifurcation dynamics for most networks in the form ACAM-J, followed by a relative stabilization during the formless ACAM-J.

Discussion

Our central research goal was to identify the mechanisms underlying minimal phenomenal experiences, here studied in the context of jhāna meditation. To address this, we fitted whole-brain models to fMRI data across the eight ACAM-J states and examined the dynamical regimes of resting-state networks, focusing in particular on the deepest stages (the formless ACAM-J and J7–J8) as representative of MPE. Our analyses revealed that the gradient of RSN dynamical regimes observed in the control condition collapsed in MPE, with most networks converging toward near-bifurcation dynamics. This convergence unfolded through distinct transition points, most prominently at J1, J4, and J5, followed by smoother changes during the formless ACAM-J. In what follows, we discuss the implications of these findings for the neuroscience of MPE and pure awareness, related theoretical frameworks, and the science of consciousness.

RSN dynamics collapse towards criticality in MPE

MPE as a critical working point

The healthy waking brain is thought to operate in an “optimal working point” near criticality, balancing sensitivity to perturbations with dynamical stability (Deco et al. 2012). Empirical work in whole-brain modeling has demonstrated this by fitting a global bifurcation parameter (homogeneous across regions) to resting-state FC, showing it to lie near the bifurcation (Deco et al. 2017). This is consistent with our genetic algorithm fitting results for the global bifurcation parameter in the counting condition, where a similar near-critical working point is observed (see Supplementary Figure S3). Crucially, however, we have seen that this global regime is sustained by a heterogeneous distribution of RSN dynamical regimes, with a few networks operating near the bifurcation while most remain further away. Indeed, the whole-brain models in the counting condition showed an important gradient in the bifurcation parameter, with the Default Mode

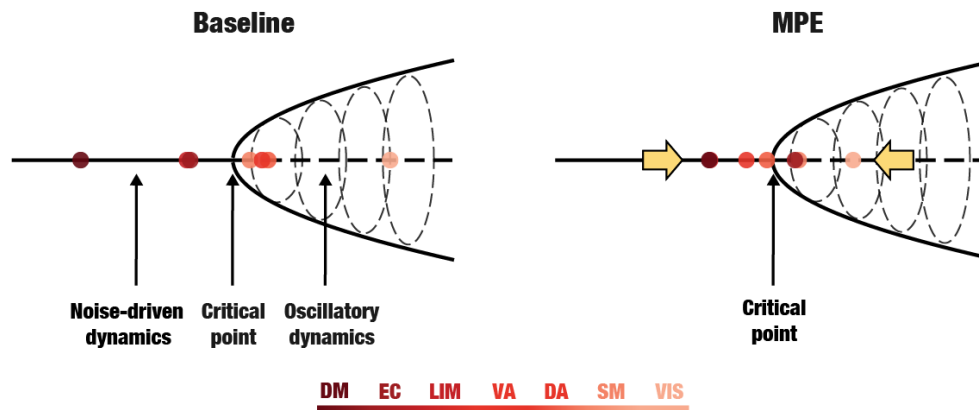


Figure 4: Brain dynamics collapse towards criticality in MPE. The gradient from noise-driven to oscillatory dynamics observed in RSN bifurcation parameters during the counting condition (left) collapses in the formless ACAM-J (right), with all networks shifting to near-critical dynamics. We hypothesize that, at baseline, the brain’s global regime is maintained through heterogeneity: different resting-state networks occupy distinct dynamical regimes, with the Default Mode and Visual network furthest from bifurcation. In contrast, during MPE, this heterogeneity collapses: all RSNs converge uniformly toward near-bifurcation dynamics. This shift suggests that MPEs reflect an alternative optimal brain state, characterized not by differentiated dynamics across RSNs, but by a uniform engagement of all networks close to criticality.

and the Visual network positioned at the extremes and furthest away from the bifurcation (see Figure 2A).

In contrast, this observable gradient collapses for the formless ACAM-J and J7–J8, which we use here as candidates for MPE. Networks that were previously far from the bifurcation point converged toward the near-bifurcation regime (see Figure 2A). This suggests that during MPE, brain regions operate uniformly close to criticality, with networks that were previously entrenched in their characteristic dynamics becoming more susceptible to the influence of the whole system. Interestingly, the global bifurcation parameter fitting results showed that the global working regime in MPE was also close to criticality (see Supplementary Figure S3). In this light, MPE appears to represent another optimal working point of the brain: one not sustained by heterogeneous dynamics across the functional gradient of RSNs but by all RSNs engaging in near-critical dynamics.

The resulting global susceptibility of networks indicates a *latent* capacity to respond to perturbations. In waking rest, RSNs are distributed across distinct dynamical regimes, but in MPE they become uniformly susceptible, suggesting a state of generalized flexibility unconstrained by hierarchical structure or entrenched patterns of activity. Importantly, this state of maximal “openness”, akin to a blank canvas, may allow the brain to reset or reorganize its dynamical repertoire, including deeply rooted attractors. Such openness could help explain why emerging from these states is often associated with radical shifts in perception, or *insights*, which are considered in contemplative traditions as central to the transformative potential of jhāna practice (Burbea 2014).

MPE criticality and related theoretical accounts

This resonates with recent computational work within the free energy principle framework, which models MPE in terms of variational free energy minimization and likewise emphasizes its relation to critical dynamics (Sandved-Smith 2024). In addition, recent mechanistic accounts frame the *jhānas* as a process of flattening hierarchical attractors, which manifests phenomenologically as a progressive *defabrication* of experience (Lopez-Sola et al. 2025). Flattening attractors in this way implies a dynamical regime where existing patterns of activity lose stability, and the system as a whole becomes more susceptible to perturbation. This description aligns closely with our finding that RSN dynamics converge toward near-critical dynamics and higher susceptibility in the formless ACAM-J.

We also described this process as a sequential disruption of the hierarchical models, which is phenomenologically described as a progressive “letting go” of increasingly basic experiences (space, consciousness, nothingness, ...) (Burbea 2014; Prest et al. 2024). Each stage involves the release of rigid priors, yielding greater sensitivity and reduced constraint, consistent with the approximation to criticality that we observe here. MPE may represent the extreme end of this spectrum of letting go, where brain dynamics come closest to the critical point.

Relatedly, recent studies using the active inference framework have proposed that the progression through the ACAM-J states and towards MPE may be driven by a systematic reduction of prior precision across the hierarchy (Laukkonen et al. 2025; Mago et al. 2024; Prest et al. 2024; Sandved-Smith 2024), also aligning with the description of a progressive flattening of hierarchical attractors. Moreover, lower precision would make the generative model more malleable, enabling the reconstruction of priors in new ways, which resonates with the transformational aspect mentioned above.

Finally, the mechanistic account of MPE we have discussed resonates with Metzinger’s interpretation of MPE as the experience of an *epistemic space* (Metzinger 2024), an “inner space holding a very large number of possibilities for knowing the world and ourselves”. It may also be compatible with Metzinger’s proposal that MPE corresponds to a representation of *tonic alertness* (Metzinger 2020). In our view, the collapse of heterogeneous RSN dynamics into a uniformly near-critical regime reflects a stripping away of hierarchical models, and what remains could be this minimal, non-egoic alertness. In this view, when object perception, self-representation, and executive control are released, the brain may settle into a globally susceptible dynamics whose neural signal corresponds to this “alertness field”.

Relationship to meditative absorption and non-dual awareness

It is important to note that our proposal of maximal susceptibility in MPE is compatible with deep meditative absorption states (Sparby et al. 2024), in which external objects are not actively represented. We hypothesize that in these states the absence of active tracking of sensory objects frees the system from the usual exogenous constraints while leaving the dynamics open. The system remains highly critical in the sense of being fully available for reconfiguration, which is precisely what is often observed upon emerging from these states.

But such absorption states are clearly not compatible with everyday functioning, given the lack of representational content. A natural question, then, is whether there are modes of experience that are maximally susceptible yet compatible with ordinary activity. A promising candidate is *nondual awareness*, which can co-occur with phenomenal content, as described in many of Metzinger’s reports (Metzinger 2024). In our recent work, we suggested that nondual awareness is not a *minimal* state, but is related to some MPEs in that awareness itself comes to the foreground of experience (Lopez-Sola et al. 2025). We proposed that this may involve an *opacification* of the modeling hierarchy, creating a distinct form of global flexibility. A provocative hypothesis is that nondual awareness also reflects a mode of near-critical brain dynamics and maximal openness, but one that remains compatible with representing other forms of phenomenal content (as opposed to deep absorption states). Testing this possibility would require applying the present modeling framework to empirical data on nondual states.

Converging empirical support

Converging evidence for the picture of MPE as near-critical states comes from recent studies on meditative practices and psychedelics using measures that reflect various signatures of criticality.

A recent study found that ACAM-J alters functional connectivity gradients, collapsing the usual hierarchical separation between sensory and higher-order regions and promoting a more globally-integrated mode (Demir et al. 2025). Such a collapse of hierarchical differentiation is consistent with our finding that MPE involves a convergence of RSN dynamics toward criticality. Additional support comes from EEG criticality studies of meditation-induced cessation events (van Lutterveld et al. 2025). These cessation events often mark the fruition of insight practice and involve profound transformations in awareness, and the fact that they also exhibit signatures of criticality strengthens the view that such dynamics may be a common marker of MPE.

A further line of evidence relates to entropy and complexity. Critical dynamics are characterized by increased complexity (Chialvo 2010), suggesting that entropy and complexity should increase progressively across the ACAM-J states and peak during MPE. Such a hypothesis was explicitly advanced in recent computational work (Mago et al. 2024), and although higher complexity and entropy have been reported in meditative states more broadly (Atad et al. 2025), specific evidence for ACAM-J is still lacking.

Psychedelics provide a further point of comparison, as they are also hypothesized to shift brain dynamics toward criticality and higher complexity (Atasoy et al. 2017; Carhart-Harris 2018; Ruffini et al. 2023; Vohryzek et al. 2025). In the REBUS framework, this shift is also interpreted as a flattening of the attractor landscape, reducing the dominance of high-level networks such as the DMN and increasing sensitivity to bottom-up signals (Carhart-Harris et al. 2019; Girn et al. 2022; Ruffini et al. 2024; Vohryzek et al. 2024a,b). This trajectory parallels the collapse of hierarchical differentiation we observed in deep ACAM-J states. This connection is especially interesting in the case of 5-MeO-DMT, which can also induce a phenomenology that resembles MPE (Timmermann et al. 2025). A recent study has shown that this compound induces atypical slow-wave dynamics in the brain (Blackburne et al. 2025), yet no study has directly examined

whether these effects involve a shift in criticality.

Together, these perspectives and empirical studies may offer complementary characterizations of the progressive shift toward criticality across ACAM-J states, a process that may culminate in an MPE state characterized by maximal openness.

Orchestrating dynamics and DMN susceptibility in MPE

Beyond the global convergence toward criticality, our results revealed shifts in which networks take on the most *orchestrating* role. Networks closer to the bifurcation exert more influence since their increased susceptibility makes them more capable of influencing system-wide dynamics. In the control counting condition, the Somatomotor network was closest to the bifurcation, suggesting a central role in shaping global activity. In MPE, however, the Dorsal Attention network assumed this role. Such a transition is consistent with the fact that ACAM-J is an attentional practice (Sparby et al. 2024), and their progression naturally involves a shift from sensorimotor to attentional mechanisms, especially into the formless stages where body and form-related perceptions fade (Brasington 2015; Burbea 2014).

In addition, DMN showed the largest change in bifurcation parameter. In the control condition, the DMN was the furthest from the bifurcation and in MPE it became markedly closer to the critical point, becoming more susceptible to perturbation. This indicates that its dynamics are no longer dominated by intrinsic activity but instead are more responsive to the system as a whole. Such a shift highlights the central role of DMN in the large-scale reorganization of brain activity during later ACAM-J states, and may reflect a transition from within- to between-network integration (Treves et al. 2025). This interpretation is consistent with empirical reports of altered DMN activity in meditation (Garrison et al. 2015), and more specifically in ACAM-J (Ganesan et al. 2024).

The progression towards MPE

Our results suggest that the collapse of hierarchical dynamics toward criticality is non-linear. Most networks approached near-bifurcation dynamics within the form ACAM-J and remained close to criticality during the transition to the formless states. At a finer scale, however, each RSN followed a distinct trajectory (Figure 3).

Interestingly, the most dramatic changes coincided with phenomenological milestones of ACAM-J (Brasington 2015; Yang et al. 2024). The first was the entry into J1, which is described as a deep absorption state characterized by intense bliss (Sparby et al. 2024). Given that this transition involves moving from a control task (counting) into meditation, a major reconfiguration in brain dynamics is to be expected, and indeed we observed such a shift. The second occurred at J4, the peak of the form ACAM-J, dominated by equanimity (Sparby et al. 2024). The third was the passage from form to formless ACAM-J (J4–J5), where the perception of form (sights, sounds, body sensations) is thought to dissolve, leaving only “boundless space” (Brasington 2015; Burbea 2014).

At each of these milestones, the largest changes were observed in the Default Mode and Visual

networks. This pattern is consistent with their position in the control condition (furthest from the bifurcation), and with our finding that the DMN undergoes the strongest overall reconfiguration. The pronounced involvement of the Visual network in the transition to the formless ACAM-J may relate to the reported loss of object perception, though we remain cautious about this interpretation.

By contrast, the formless ACAM-J were not marked by large, discrete changes. The DMN and SM networks continued to show modest but consistent shifts toward criticality, while most other networks remained stably near the bifurcation (see Supplementary Figure S2). The absence of large reconfigurations in the formless ACAM-J may reflect their phenomenological character as more pacified states, in which transitions are experienced as slower and more gradual (Brasington 2015). This pattern is also consistent with recent accounts (Lopez-Sola et al. 2025), in which the progression through the formless ACAM-J corresponds to the flattening of increasingly deep and subtle attractors. In this view, the formless ACAM-J consolidate the system into a broadly flattened dynamical landscape, with residual network heterogeneity, rather than driving further large-scale reconfigurations.

This nonlinear collapse of network dynamics toward near-critical regimes may reflect how the brain endogenously accesses MPE through meditation. The transition toward a globally susceptible state may be constrained by biological dynamics, such that networks cannot shift linearly toward bifurcation together but must do so in stages. Whether other pathways to MPE, such as psychedelic states, follow a similar stepwise progression remains an open question.

Implications for consciousness research

The interest for MPE was originally motivated by the search for a minimal model of consciousness, providing a framework to study what the most basic form of conscious awareness might look like (Metzinger 2020). Our findings suggest that deep ACAM-J states, as serious candidates for MPE, may correspond to a global collapse of hierarchical differentiation into a near-critical regime marked by maximal susceptibility and long-range spatio-temporal correlations. This carries several implications for theories of consciousness.

First, the convergence of RSN dynamics toward criticality in MPE is consistent with theories that emphasize global accessibility and integration as core features of conscious states. For instance, the Global Workspace Theory proposes that consciousness arises from widespread ignition or broadcast across the brain (Baars 2005). A uniformly near-critical regime would naturally support such ignition, since the entire system is highly susceptible to perturbations. In this light, MPE could be understood as a configuration with maximal *capacity* for broadcast, even if little or nothing is actually broadcast. Similarly, Integrated Information Theory highlights integration as a hallmark of consciousness (Tononi 2008); our results suggest that MPE may represent a configuration in which integration is achieved not through hierarchical specialization between RSNs but through increased spatio-temporal correlation across networks.

Second, the global collapse we observed during MPE offers an interesting perspective on its status as a minimal model. If consciousness in its most stripped-down form is associated with an undifferentiated, maximally critical regime, this may point toward a primordial mode of

conscious functioning. From a developmental and evolutionary perspective, early forms of consciousness in fetuses, infants, or non-human animals might resemble such globally flexible, weakly differentiated states, before rigid hierarchical control is established. Evidence for this comes from developmental neuroimaging, which shows that functional connectivity in fetuses and newborns is dominated by global, low-dimensional patterns, with higher-order network differentiation only emerging later in infancy and childhood (Gao et al. 2017). Comparative research also suggests that simpler organisms and non-human animals often display globally integrated architectures, lacking the strong hierarchical segregation characteristic of adult human brains (Van den Heuvel et al. 2016).

This perspective highlights the transformative potential of MPE. In ordinary functioning, hierarchies and rigid dynamics constrain the system, ensuring survival and goal-directed behavior but limiting flexibility. Through practices such as ACAM-J, these constraints can be temporarily released, allowing the system to return to a globally open regime. Such reversals may explain why some minimal experiences, such as pure awareness, are so often described as transformative.

Limitations and future work

We chose to parametrize the whole-brain models in terms of resting-state networks. This was a natural first step, given their robustness and frequent use in both empirical and modeling studies. However, alternative priors could be equally or more informative. For example, functional gradients might capture cortical hierarchical organization, and neurotransmitter receptor maps could link model parameters more directly to neuromodulatory dynamics. Future work should explore these alternative parametrizations, as they may reveal additional insights into the mechanisms of MPE.

In the main text, we used the formless ACAM-J as candidate MPE states. While one might argue that these encompass some non-minimal stages, we verified in the Supplementary Information that our results remain robust when restricting the analysis to J7–J8, which are arguably the closest to minimal (Lopez-Sola et al. 2025). A further limitation concerns the diversity of jhāna practice and phenomenology. Although the eightfold sequence is widely recognized, descriptions vary considerably across traditions, teachers, and individual practitioners (Sparby et al. 2024). Our use of the last ACAM-J stages as representative of MPE states provides a useful operationalization, but they only provide one example of possible minimal phenomenal experiences. As proposed in previous work (Laukkonen et al. 2023; Lopez-Sola et al. 2025), extended cessations, sometimes reported to occur after J8, may also provide unique insights into the neurobiological mechanisms behind MPE and could be studied using similar modeling pipelines.

The progression through ACAM-J itself also presents challenges. With eight distinct stages and subtle phenomenological shifts between them, disentangling the precise neural dynamics that underlie the trajectory is difficult, both in terms of modeling and first-person description. Our analysis highlights several key transitions, but much remains to be explored in terms of the fine-grained evolution of network dynamics across the entire sequence. Future studies could combine

our modeling approach with neurophenomenological designs that include moment-to-moment reports (Timmermann et al. 2023).

The Hopf model provides a well-established framework for studying whole-brain dynamics, and the bifurcation parameter can be related to notions like criticality. To further understand the underlying neurophysiology of cortical layers or neurotransmitter systems, future work could use more realistic models, such as neural mass models, to compliment our interpretation by linking parameter changes more closely to neurobiological mechanisms. For example, recent work has developed a laminar neural mass model that implements prediction error computation (Ruffini et al. 2025) and has been used in whole-brain model simulations of psychedelic effects (Gendra et al. 2024). Extending this approach to advanced meditation and MPE could yield a more neurobiologically grounded account of our findings.

Finally, the methods developed here could be applied to other candidates for minimal phenomenal experiences. Psychedelics such as 5-MeO-DMT, which have been associated with MPE-like phenomenology (Timmermann et al. 2025), or states described in contemplative traditions such as Clear Light sleep (Alcaraz-Sánchez 2024), may provide important points of comparison. Applying the same modeling framework across these different pathways to minimal experience could help identify both shared and distinct neurobiological mechanisms.

Conclusions

In conclusion, our study provides unique insights into the neuroscience of pure awareness and MPE, drawing on advanced meditation and going beyond abstract modeling and analyses of functional data to model the underlying mechanisms and dynamical regimes. In doing so, it provides the kind of formal modeling needed to bridge first-person phenomenology with third-person neural data, and avoids the pitfalls of phenomenologically-inspired “mathematical storytelling”, relating formal mechanisms to testable neurobiological hypotheses. As one of the first whole-brain modeling studies of meditative absorption practice and MPE, this work aims to set a foundation for a mechanistic understanding of advanced meditation and to inspire further integration of contemplative phenomenology with computational neuroscience.

Contributions

J.V. and E.L-S. conceived the modeling approach and study reported here. M.D.S. conceived of and funded the broader study. J.V. and E.L-S. designed and performed the analyses, with methodological input from Y.S-P. and G.D.. W.F.Z.Y, T.S., and M.D.S. collected the meditation dataset. W.F.Z.Y. handled and preprocessed the fMRI data. J.V., Y.S-P, and G.D. developed and implemented the whole-brain modeling framework, including optimization procedures. M.D.S. and G.D. provided conceptual guidance and supervised the work. J.V. and E.L-S. wrote the first draft of the manuscript. All authors contributed to the interpretation of results, manuscript revisions, and approved the final version.

Conflict of Interest

G.R. and E.L-S. work for Neuroelectrics, a company developing computational brain stimulation solutions for neuropsychiatric disorders.

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